Algebraic Geometry Notes

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1 Affine Algebraic Varieties

Definition 1.1. Let K be a field. Then an Affine Algebraic Variety is a set of the form

$$\mathcal{V}(I) = \{(a_1, a_2, \ldots, a_n) \in K^n \mid f(a_1, a_2, \ldots, a_n) = 0, \forall f \in I\},$$

where I is an ideal $I \lhd K[x_1, x_2, \ldots, x_n]$.

Proposition 1.2. Let I_1, I_2 be ideals of $K[x_1, \ldots, x_n]$ be such that $I_1 \subseteq I_2$, then $\mathcal{V}(I_1) \supseteq \mathcal{V}(I_2)$.

Example 1.3.

(i) If $I = \{0\}$, then $\mathcal{V}(I) = K^n$.

(ii) If $I = K[x_1, x_2, \ldots, x_n]$, then $\mathcal{V}(I) = \{\emptyset\}$.

(iii) If $I = \langle x^2 + y^2 - 1\rangle \lhd \mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the circle of radius 1 centred at the origin defined by the equation $x^2 + y^2 = 1$ in $\mathbb{R}^2$.

(iv) Let $f_1, f_2, \ldots, f_r$ be polynomials of degree 1 in $x_1, x_2, \ldots, x_n$, and let $I = \langle f_1, f_2, \ldots, f_r\rangle$ be the ideal generated byu them. Then $\mathcal{V}(I)$ is the set of solutions of the system of linear equations $f_1 = f_2 = \cdots = f_r = 0$. Such a variety is called an affine subspace of K^n. This, in particular, includes sets containing a single point. If $P = (a_1, a_2, \ldots, a_n) \in K^n$, then the ideal $I = \langle x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n\rangle$ defines the variety $\mathcal{V}(I) = \{P\}$.

(v) Let $I = \langle y^2 - x^3\rangle \lhd \mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabol, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabol, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabola.

(vi) Let $I = \langle y^2 - x^3 - x^2\rangle\mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the nodal cubic curve defined by the equation $y^2 = x^3 + x^2$.

(vii) If $I = \langle x, y\rangle \lhd [x, y]$, then $\mathcal{V}(I) = \{(0, 0)\} \subset \mathbb{R}^2$.

(viii) If $I = \langle x^2, y^2\rangle \lhd [x, y]$, then $\mathcal{V}(I) = \{(0, 0)\} \subset \mathbb{R}^2$.

(ix) If $I = \left\langle x^2 + y^2 + z^2 - 1, x^2 - x + y^2\right\rangle \lhd \mathbb{R}[x, y, z]$, then $\mathcal{V}(I) \subset \mathbb{R}^3$ is a figure of eight curve from an intersection of the cylinder, $y^2 = x^2 - x$, and the sphere of radius 1 at the origin.

(x) The special linear group $SL(n\mathbb{R})$ is also an affine algebraic variety. The set of all $n \times n$ real matrices is $\mathbb{R}^{n^2}$, the determinant is a polynomial in the entries of the matrix, and therefore $SL(n, \mathbb{R}) = \mathcal{V}(\langle \det(A) - 1\rangle)$.

Proposition 1.4. Let K be a field. Let $f_1, f_2, \dots, f_r \in K[x_1, x_2, \dots, x_n]$ and let $I = \langle f_1, f_2, \dots, f_r \rangle$. Then

$$\mathcal{V}(I) = \{(a_1, a_2, \dots, a_n) \in K^n \mid f_i(a_1, a_2, \dots, a_n) = 0, \forall 1 \leq i \leq r\}.$$

Proof. We know $f_i \in I$ for each i, so if $(a_1, \dots, a_n) \in \mathcal{V}(I)$, then $f_i(a_1, \dots, a_n) = 0$ for every i. Conversely, assume that $f_i(a_1, \dots, a_n) = 0$ for every i. Any $f \in I$ can be written as a $K[x_1, \dots, x_n]$-linear combination of the f_i, so

$$f(a_1, \dots, a_n) = \sum_{i=1}^{r} f_i(a_1, \dots, a_n) g_i(a_1, \dots, a_n) = 0,$$

so $(a_1, \dots, a_n)$. $\square$

Definition 1.5. A ring R is called Noetherian if all its ideals can be generated by finitely many elements.

Corollary 1.6 (Hilbert Basis Theorem). If K is a field, then the polynomial ring $K[x_1, \dots, x_n]$ is Noetherian for any $n \geq 0$.

Example 1.7.

(i) If R is Noetherian and $I \lhd R$, then R/I is also Noetherian.

(ii) $\mathbb{Z}$, the ring of integers, is Noetherian because it is a PID, so every ideal is generated by one element.

(iii) $K[x_1, x_2, \dots]$, the polynomial rinf over the field K in finitely many variables is an example of a non-Noetherian ring, because the ideal $\langle x_1, x_2, \dots \rangle$ is not finitely generated.

Proposition 1.8.

(i) Let $V_1 = \mathcal{V}(I_1), \dots, V_k = \mathcal{V}(I_k)$ be affine algebraic varieties in K^n. Then

$$V_1 \cup \dots \cup V_k = \mathcal{V}(I_1 \cap \dots \cap I_k) = \mathcal{V}(I_1 I_2 \cdots I_k)$$

is also an affine algebraic variety.

(ii) Let $V_\alpha = \mathcal{V}(I_\alpha), \alpha \in A$ be affine algebraic varieties in K^n. Then

$$\bigcap_{\alpha \in A} V_\alpha = \mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right)$$

is also an affine algebraic variety.

(iii) Let $V_1 = \mathcal{V}(I_1) \subseteq K^m, V_2 = \mathcal{V}(I_2) \subseteq K^n$ be affine algebraic varieties. Then $V_1 \times V_2 \subseteq K^m \times K^n \cong K^{m+n}$ is also an affine algebraic variety.

Proof.

(i) We first assume that $k = 2$. Let $P \in V_1 \cup V_2$ and let $f \in I_1 \cap I_2$. If $P \in V_1$, then $f(P) = 0$ because $f \in I_1$, and similarly if $P \in V_2$ then $f(P) = 0$ because $f \in I_2$. Therefore in either case we have $f(P) = 0$, so $P \in \mathcal{V}(1_1 \cap I_2)$, so $V_1 \cup V_2 \subseteq \mathcal{V}(I_1 \cap I_2)$.
We also know $I_1 I_2 \subseteq I_1 \cap I_2$, so we also have $\mathcal{V}(I_1 \cap I_2) \subseteq \mathcal{V}(I_1 I_2)$.
Let now $P \in K^n \setminus (V_1 \cup V_2)$. Then $P \notin V_1$, so there exists $f_1 \in I_1$ such that $f_1(P) \neq 0$. Similarly, $P \neq V_2$ so there exists $f_2 \in I_2$ such that $f_2(P) \neq 0$. Then $f_1 f_2 \in I_1 I_2$ and $(f_1 f_2)(P) = f_1(P) f_2(P) \neq 0$, therefore $P \notin \mathcal{V}(I_1 I_2)$. This implies $\mathcal{V}(I_1 I_2) \subseteq V_1 \cup V_2$. Hence we obtain

$$V_1 \cup V_2 = \mathcal{V}(I_1 \cap I_2) = \mathcal{V}(I_1 I_2).$$

For $k > 2$ we use induction on k.

(ii) Let $P \in \bigcap_{\alpha \in A} V_\alpha$ and let $f = \sum_{\alpha \in A} I_\alpha$. Then f can be written as

$$f = \sum_{\alpha \in A} f_\alpha$$

with $f_\alpha \in I_\alpha$ for every $\alpha \in A$ and $f_\alpha = 0$ for all but finitely many α. We know $P \in V_\alpha$ for every $\alpha \in A$, so $f_\alpha(P) = 0$ for every $\alpha \in A$, therefore

$$f(P) = \sum_{\alpha \in A} f_\alpha(P) = \sum_{\alpha \in A} 0 = 0.$$

Hence $P \in \mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right)$. This implies

$$\bigcap_{\alpha \in A} V_\alpha \subseteq \mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right).$$

Assume now that $P \in K^n \setminus \bigcap_{\alpha \in A} V_\alpha$. Then there exists $\alpha_0 \in A$ such that $P \notin V_{\alpha_0}$, therefore there exists $f \in I_{\alpha_0}$ such that $f(P) \neq 0$. Now $f \in \sum_{\alpha \in A} I_\alpha$, therefore $P \notin \mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right)$. This implies that

$$\mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right) \subseteq \bigcap_{\alpha \in A} V_\alpha.$$

(iii) Let $x_1, \ldots, x_m$ be coordinates on K^m, and $y_1, \ldots, y_n$ be coordinates on K^n. Let $J_1 = \langle I_1 \rangle \lhd K[x_1, \ldots, x_m, y_1, \ldots, y_n]$, where I_1 is an ideal in

$K[x_1, \ldots, x_m]$. We claim that $\mathcal{V}(J_1) = V_1 \times K^n$.
Let $P = (a_1, \ldots, a_m, b_1, \ldots, b_n) \in \mathcal{V}(J_1)$. Then $f(P) = 0$ for every $f \in I_1$, since $I_1 \subset J_1$. As f only involves the variables $x_1, \ldots, x_m$,

$$0 = f(P) = f(a_1, \ldots, a_m, b_1, \ldots, b_n) = f(a_1, \ldots, a_m).$$

Hence $(a_1, \ldots, a_m) \in \mathcal{V}(I_1) = V_1$, so $P \in V_1 \times K^n$, therefore $\mathcal{V}(J_1) \subseteq V_1 \times K^n$.
Now let $P = (a_1, \ldots, a_m, b_1, \ldots, b_n) \in V_1 \times K^n$. Let $f \in J_1$. Then f can be written as $f = \sum_{i=1}^{r} f_i g_i$, where $f_i \in I$ and $g_i \in K[x_1, \ldots, x_m, y_1, \ldots, y_n]$. Now

$$f_i(P) = f_i(a_1, \ldots, a_m, b_1, \ldots, b_n) = f_i(a_1, \ldots, a_m) = 0.$$

Therefore $f(P) = 0$, so $P \in \mathcal{V}(J_1)$. This means $V_1 \times K^n \subseteq \mathcal{V}(J_1)$, and so we have shown that $\mathcal{V}(J_1) = V_1 \times K^n$. Similarly, if we define $J_2 = \langle I_2 \rangle \lhd K[x_1, \ldots, x_m, y_1, \ldots, y_n]$, then $\mathcal{V}(J_2) = K^m \times V_2$. Hence

$$V_1 \times V_2 = (V_1 \times K^n) \cap (K^m \times V_2) = \mathcal{V}(J_1) \cap \mathcal{V}(J_2) = \mathcal{V}(J_1 + J_2). \quad \square$$

Example 1.9.

(i) We wish to find an ideal $I \lhd \mathbb{R}[x, y]$ such that $\mathcal{V}(I) = (1, 1), (2, -3)$. Let $I_1 = \langle x - 1, y - 1 \rangle$ and $I_2 = \langle x - 2, y + 3 \rangle$. Then $\mathcal{V}((\,) I_1) = (1, 1)$ and $\mathcal{V}(I_2) = (2, -3)$. Hence, by Proposition 1.8(i),

$$I = \mathcal{V}((x-1)(x-2), (x-1)(y+3), (y-1)(x-2), (y-1)(y+3))$$

is a suitable ideal.

(ii) Let $V_1 = \mathcal{V}\left(\left\langle x_1^2 + x_2^2 - 1 \right\rangle\right) \subset \mathbb{R}^2$ and let $V_2 = \mathcal{V}(\langle y_1 - y_2 \rangle) \subset \mathbb{R}^2$. Then $V_1 \times V_2 = \mathcal{V}\left(\left\langle x_1^2 + x_2^2 - 1, y_1 - y_2 \right\rangle\right) \subset \mathbb{R}^4$ is a cylinder.

Definition 1.10. Let K be a field and let $V \subseteq K^n$ be an affine algebraic variety. W is a subvariety of V if $W \subseteq V$ and W is also an affine algebraic variety.

1.1 Affine Spaces

Definition 1.11. Let K be a field, and let V be a finite dimensional vector space over K. Then $U \subseteq V$ is called an affine subspace of V if $U = \emptyset$ or or $U = u_0 + W = \{u_0 + w \mid w \in W\}$, where $u_0 \in U$ and W is a linear subspace of V.

Example 1.12.

(i) $K^n, \emptyset$, and any set containing a single point are affine subspaces of K^n.

(ii) The line $y = \frac{1}{2}x + 1$ is an affine subspace of $\mathbb{R}^2$, in this case we can take $W = span(2, 1)$, which is the line $y = \frac{1}{2}x$, and $u_0 = (0, 1)$.

Proposition 1.13. Let $U = u_0 + W$ be a non-empty affine subspace in a vector space V. Then the linear subspace W is uniquely deteremined by U, and U can be written as $U = u_1 + W$ for any $u_1 \in U$.

Proof. By definition, $U = u_0 + W$, and therefore $W = U - u_0 = \{u - u_0 \mid u \in U\}$. Let $u_1 \in U$ be arbitrary, then $u = u_0 + w_1 \in W$. We consider the set

$$U - u_1 = \{u - u_1 \mid u \in U\} = \{u - u_0 - w_1 \mid u \in U\}.$$

We have $u - u_0 \in U - u_0 = W$, therefore $u - u_0 - w_1 \in W$, as W is a linear subspace. This means that $U - u_1 \subseteq W$. Let $w \in W$, then

$$w + u_1 = w + w_1 + u_0 = (w + w_1) + u_0 \in U,$$

so $w \in U - u_1$, therefore $U - u_1 = W$. Hence $U = u_1 + W$ for any $u_0 \in U$. $\square$

Proposition 1.14. Let U_1, U_2 be affine subspaces of V. Then $U_1 \cap U_2$ and $U_1 + U_2 = \{u_1 + u_2 \mid u_1 \in U_1, u_2 \in U_2\}$ are also affine subspaces of V.

Proof. If $U_1 \cap U_2 = \emptyset$, then $U_1 \cap U_2$ is an affine subspace, so assume that $U_1 \cap U_2 \neq \emptyset$ and let $u_0 \in U_1 \cap U_2$. So $U_1 = u_0 + W_1, U_2 = u_0 + W_2$, where W_1, W_2 are vector subspaces of V and $U_1 \cap U_2 = u_0 + (W_1 \cap W_2)$. THe intersection of linear subspaces is also a linear subspace, therefore $U_1 \cap U_2$ is an affine subspace of V.
If either $U_1 = \emptyset$ or $U_2 = \emptyset$, then $U_1 + U_2 = \emptyset$, so it too is an affine subspace of V.
If neither U_1 nor U_2 is the empty set, then we can write them as $U_1 = v_1 + W_1, U_2 = v_2 + W_2$, where $v_1 \in U_1$ and $v_2 \in U_2$ and W_1, W_2 are linear subspaces of V. We have

$$\begin{aligned} U_1 + U_2 = &\{u_1 + u_2 \mid u_1 \in U_1, u_2 \in U_2\} \\ &\{(v_1 + w_1) + (v_2 + w_2) \mid w_1 \in W_1, w_2 \in W_2\} \\ &\{(v_1 + v_2) + w \mid w \in W_1 + W_2\}. \end{aligned}$$

$W_1 + W_2$ is also a linear subspace of V, therefore $U_1 + U_2$ is an affine subspace. $\square$

Definition 1.15. The dimension of an affine subspace $U = u_0 + W$ is defined to be $\dim W$. For convention we write $\dim \emptyset = -1$.

Definition 1.16. Let K be a field. K^n, considered as an affine subspace of itself, is called an n-dimensionsal affine space over K, and is denoted by $\mathbb{A}^n(K)$ or $\mathbb{A}^n$ if the field is understood.

Definition 1.17. Let V_1, V_2 be vector spaces. An affine map from V_1 to V_2 is defined to be a function $\varphi : V_1 \to V_2$ which can ve written in the form $\varphi(x) : V_1 \to V_2$ is a linear transformation and $b \in V_2$.

Definition 1.18. Let X, Y be subsets of a vector space V. X and Y are called affine equivalent if there exists mutually inverse affine maps $\varphi, \psi : V \to V$ such that $\varphi(X) = Y$ and $\psi(Y) = X$.